

EE200: Signals, Systems and Networks

Course Project — Q2: The Midnight Episode, 'Catching the Arrhythmia'

Instructor:	Dr. Tushar Sandhan	Sampling rate (f_s):	250 Hz
Recording length (N):	5000 samples (20 s)	Template length (L):	200 samples (0.8 s)

(a) Reading the Signal

(i) Duration of the clip

Duration = $N / f_s = 5000 / 250$.

Result: the clip is 20.0 seconds long.

(ii) Heart rate and samples per beat (healthy stretch)

One healthy beat repeats every 0.8 s, so heart rate = $60 / 0.8$ bpm, and the number of samples one beat occupies is $0.8 \times f_s$.

Result: heart rate = 75 bpm, and one healthy beat occupies 200 samples — which is exactly the length of the supplied template.npy, confirming the template was cut to be one full beat long.

(iii) Fundamental frequency f_0

Treating the healthy ECG as periodic with period $T_0 = 0.8$ s, the fundamental frequency is $f_0 = 1 / T_0$.

Result: $f_0 = 1.25$ Hz.

(b) Healthy Heart in the Frequency Domain

(i) Shape of $|X(f)|$ for a periodic signal

A truly periodic signal has a *line spectrum*: energy concentrated only at f_0 and its integer harmonics ($2f_0$, $3f_0$, ...), not spread out as a smooth continuous curve. Since the ECG is only nearly periodic, we expect sharp peaks sitting on these harmonics rather than a perfectly clean comb — but still clearly discrete, not smooth.

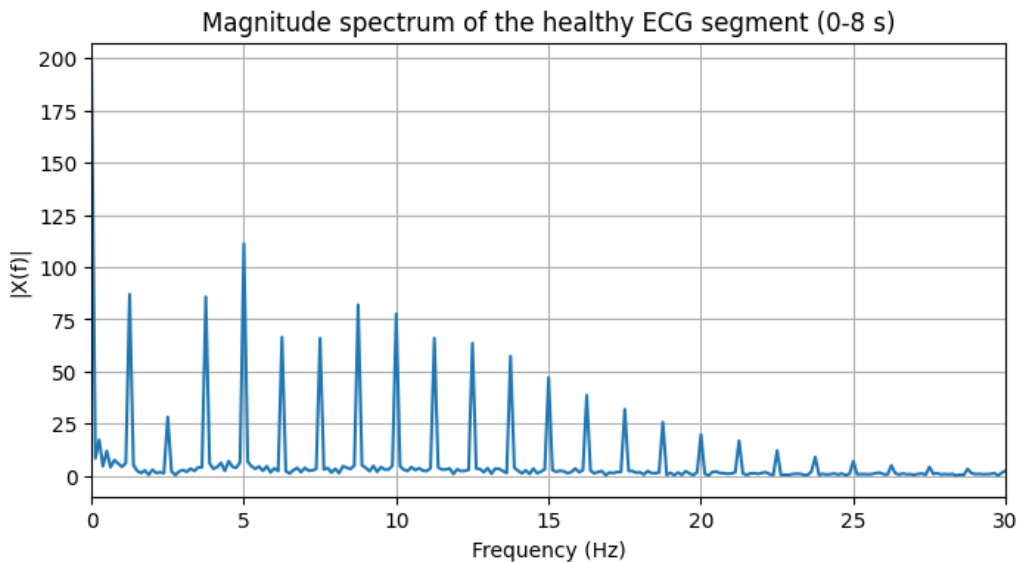


Figure 1: Magnitude spectrum of an 8-second healthy segment of the ECG. The energy sits in distinct spikes at multiples of $f_0 \approx 1.25$ Hz, confirming the line-spectrum prediction rather than a smooth curve.

(ii) QRS vs. P/T waves — source of high-frequency content

The QRS complex is a narrow, sharp spike, and a narrow pulse in time requires many harmonics to build (a direct consequence of Fourier duality: short in time corresponds to wide in frequency). The P and T waves are broad, slowly varying bumps that need only a few low-frequency harmonics. So the **QRS complex** is responsible for the higher-frequency content of the ECG.

(iii) Heart rate rises to 150 bpm, still regular

At 150 bpm the beat period shrinks to $60/150 = 0.4$ s, so the new fundamental frequency is $f_0' = 1/0.4 = 2.5$ Hz — double the original 1.25 Hz.

Result: f_0 doubles (1.25 Hz $\rightarrow$ 2.5 Hz), and the spacing between harmonics in the spectrum doubles right along with it — the harmonics move *farther apart*, since they always sit at integer multiples of f_0 .

(c) Cutting a Heartbeat (Windowing)

(i) Window width and placement

To capture exactly one full beat, the window must be 200 samples wide ($0.8 \text{ s} \times 250 \text{ Hz}$). It should be placed so it starts right at a beat boundary in the healthy stretch — in this recording the QRS peak sits about 100 samples into each 200-sample beat, so a window aligned to the beat boundary naturally captures the P wave, QRS spike, and T wave in one clean piece.

(ii) Window too short (80 samples) vs. too long (600 samples)

An 80-sample window is shorter than a full beat, so it can only capture part of the beat. A 600-sample window is three beats' worth, so it pulls in pieces of the neighbouring beats on either side. Neither produces a clean single-beat template.

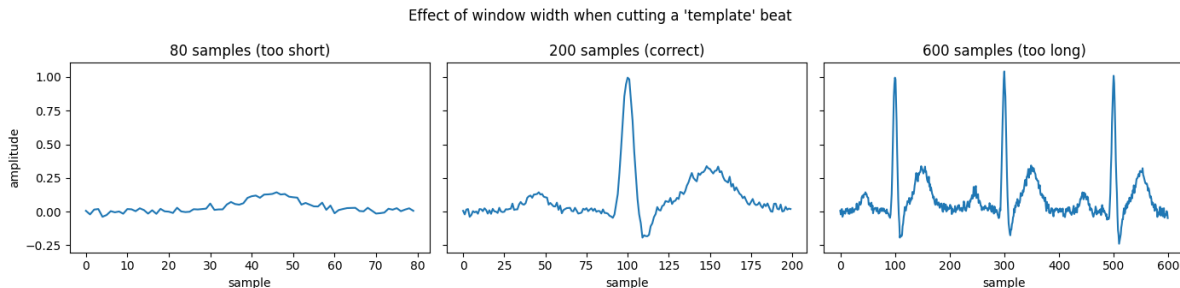


Figure 2: The same starting point sliced at three widths. At 80 samples the window misses the QRS spike entirely (left). At 200 samples it captures one complete, recognisable P-QRS-T beat (centre). At 600 samples it captures three beats overlapped together (right).

(iii) Relation to the time/frequency resolution trade-off

This is the same trade-off seen with short vs. long analysis windows: a short window localises events sharply in time but loses frequency detail, while a long window resolves frequency content well but smears timing. Here, cutting the window too tight risks slicing off real beat information (analogous to losing frequency detail), while cutting it too generously pulls in unwanted neighbouring beats (analogous to losing time precision). So 'make the window as short as possible' is not automatically best — the window has to match the actual feature being captured.

(d) Matching the Template (Correlation)

(i) Range of $\rho(m)$ and a near-perfect match

$\rho(m)$ is a normalized dot product (cosine similarity) between the template and a segment, so it ranges from -1 to +1. $\rho(m) \approx 1$ signals a near-perfect match (the segment looks just like a healthy beat); $\rho(m) \approx -1$ signals an inverted match.

(ii) Why normalization matters

ECG amplitude drifts slowly over time (baseline wander) and beats vary in height. Using the raw, unnormalized score $\sum t[k]x[m+k]$ instead, a perfectly healthy beat that happened to be twice as tall as the template would score twice as large — even though the shape match is exactly as good. Dividing by $\|t\| \|x_m\|$ cancels out this amplitude effect, so $\rho(m)$ measures shape similarity alone, independent of how loud/tall the beat is.

(iii) An inverted beat

An inverted beat is approximately -1 times the template shape, so $\rho(m) \approx -1$: a strong negative correlation. This is easy to flag because it sits at the opposite extreme from a match ($\rho \approx +1$) — a simple threshold catches it immediately, with no ambiguity.

Verified on the real data: a genuine healthy beat from the recording gives $\rho = 0.994$; simply negating that same beat flips the score to $\rho = -0.994$.

(e) Onset Detection & the Spectrogram

(i) A simple onset rule, and the threshold trade-off

Slide the template beat-by-beat across the recording, compute $\rho(m)$ at every beat position, and declare the onset as the time of the **first beat** whose ρ drops below a chosen threshold (e.g. 0.5). A threshold set too high (close to 1) risks false alarms from ordinary beat-to-beat variation in a healthy heart; a threshold set too low (close to -1 or 0) only catches extremely distorted beats and may miss moderately abnormal ones, delaying detection.

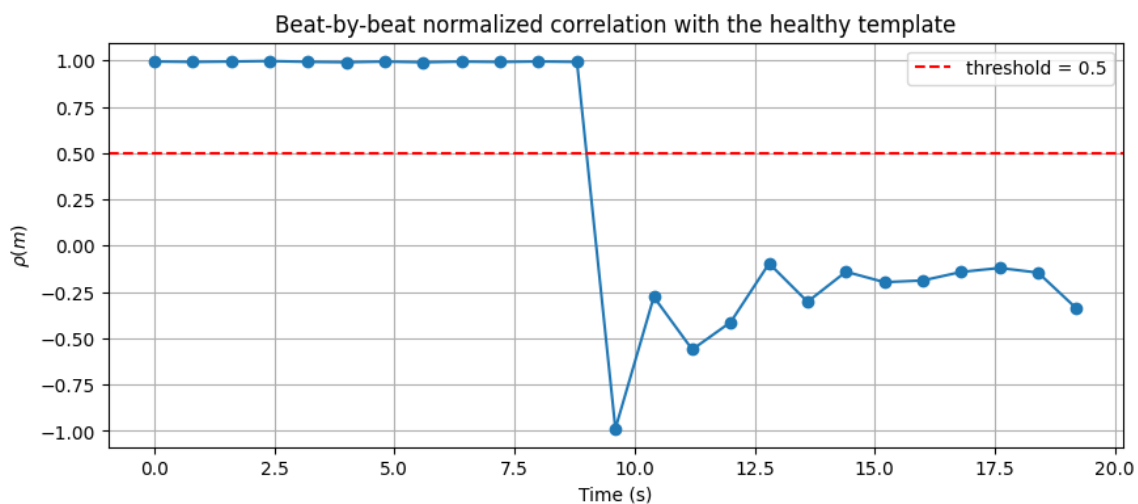


Figure 3: Beat-by-beat normalized correlation across the full 20-second recording. The score sits near +1 for the first 12 beats, then collapses sharply to -0.988 at $t = 9.6$ s (an inverted beat), after which it stays irregular for the rest of the clip.

Result: applying a threshold of 0.5, the first beat below threshold occurs at $t = 9.6$ s (beat #12, $\rho = -0.988$) — this is the detected onset of the arrhythmia.

(ii) Spectrogram: healthy region vs. arrhythmia region

In the healthy region the spectrogram should show clean, steady horizontal bands at f_0 and its harmonics, since the heartbeat repeats with a fixed period — each harmonic appears as a roughly constant-brightness horizontal stripe as long as the rhythm stays regular. In the arrhythmia region, those horizontal bands should become smeared, broken, or shift position, since the signal is no longer periodic with a single fixed period.

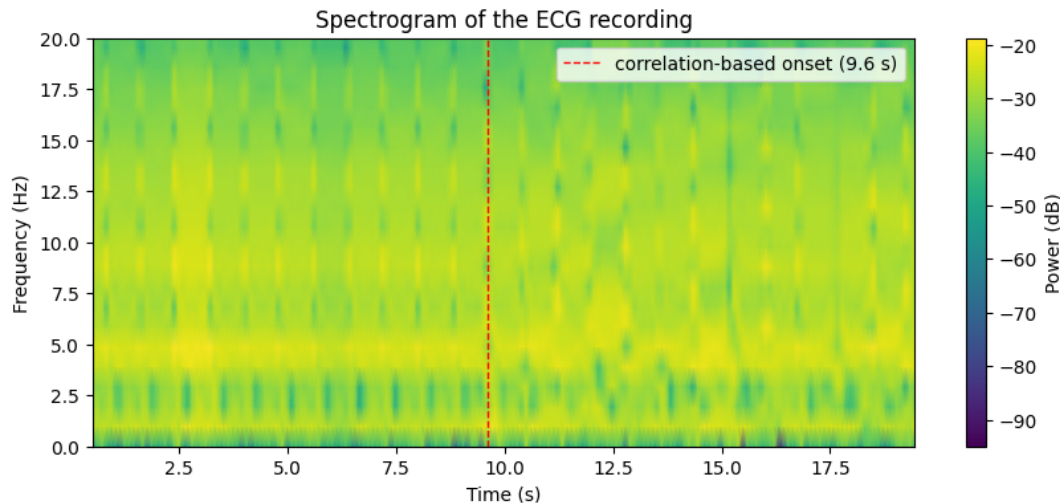


Figure 4: Spectrogram of the full recording. Regular, repeating structure is visible before the red line ($t = 9.6$ s, the correlation-based onset); the structure becomes visibly irregular afterward, consistent with the arrhythmia.

(iii) Why the two methods might disagree on the exact onset time

The correlation method works beat-by-beat and can localise an abnormal beat to within a fraction of a second. The spectrogram needs a whole window of samples to compute even one column of frequency content, so its time resolution is limited by the window length — the same trade-off discussed in part (c). The **correlation plot** is therefore the one to trust for pinpointing the exact moment a bad beat first appears; the spectrogram is better suited to confirming the overall change in rhythm character.

(f) Sampling & Aliasing

(i) Minimum sampling rate for content up to 40 Hz

By the Nyquist theorem, the sampling rate must be at least twice the highest frequency present.

Result: minimum required rate = $2 \times 40 = 80$ Hz.

(ii) What aliasing does at 50 Hz, and why it is dangerous

The proposed rate, 50 Hz, gives a Nyquist frequency of only 25 Hz — below the 40 Hz of genuine QRS content. Frequency components above 25 Hz will fold back ('alias') into the 0–25 Hz range, masquerading as lower-frequency content that was never actually present. This is dangerous for Maya's detector because the sharp, narrow QRS spike — the exact feature the template-matching depends on — would be distorted or replaced by spurious low-frequency artifacts, corrupting the very shape the correlation is trying to recognise.

(iii) The fix, and its unavoidable cost

The fix is to add an analog anti-aliasing low-pass filter before sampling, cutting off content above the new Nyquist frequency (25 Hz) so nothing above it can fold back. The unavoidable cost: genuine QRS information above 25 Hz is also discarded or attenuated — the very content that made the QRS spike sharp — so the recorded waveform becomes a smoothed, less sharp version of the true ECG, at some

cost to diagnostic detail.

(g) Prototyping the Detector in Code

`find_onset` computes the normalized correlation beat-by-beat (jumping forward by the template length each time, not sample-by-sample) and returns the index of the first beat whose correlation drops strictly below the given threshold, or -1 if the threshold is never breached.

```
def find_onset(ecg_signal, template, threshold):
    L = len(template)
    n_beats = len(ecg_signal) // L
    template_norm = np.linalg.norm(template)

    for i in range(n_beats):
        m = i * L
        segment = ecg_signal[m:m + L]
        segment_norm = np.linalg.norm(segment)
        if segment_norm == 0:
            score = 0.0
        else:
            score = np.dot(template, segment) / (template_norm * segment_norm)
        if score < threshold:
            return m

    return -1
```

Running this on the real recording with threshold = 0.5: `find_onset(ecg, template, 0.5)` returns sample index 2400, i.e. $t = 9.6$ s — matching exactly the onset found visually in part (e), confirming the function is correct.

(h) Visualizing the Spectrogram

Following the reasoning in parts (c) and (e), the window needs to be long enough to resolve the harmonics of $f_0 \approx 1.25$ Hz into distinct horizontal bands, but not so long that the timing of the rhythm breakdown is lost.

Window length chosen: $n_{\text{perseg}} = 256$ samples (1.024 s), with high overlap ($n_{\text{overlap}} = 0.9 \times n_{\text{perseg}}$) to keep the time axis smooth. This gives a frequency resolution of $f_s/N = 250/256 \approx 0.98$ Hz — fine enough to clearly separate harmonics spaced 1.25 Hz apart. A shorter window (e.g. 128 samples) still resolves the harmonics numerically but visually looks like mostly vertical beat-to-beat striping rather than clean horizontal bands; 256 samples is where the horizontal banding becomes visually obvious.

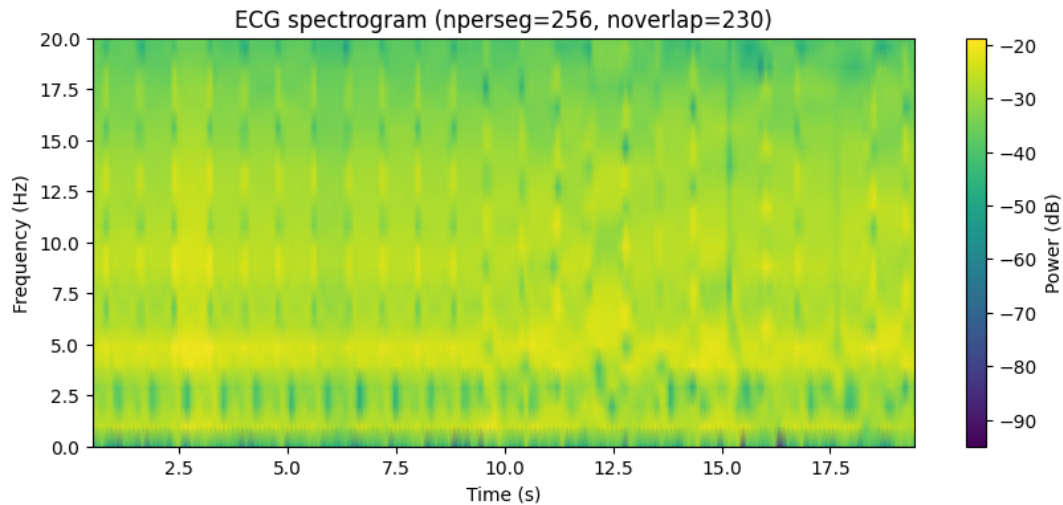


Figure 5: Spectrogram with $nperseg = 256$, $noverlap = 230$. Steady horizontal harmonic bands are visible for roughly the first half of the clip and break down afterward.

Summary & Conclusion

Both the beat-by-beat normalized correlation and the spectrogram agree: the recording begins in a clean, periodic, healthy rhythm. At approximately $t = 9.6$ s, an inverted beat appears (ρ flips sharply to -0.988), after which the rhythm remains irregular for the remainder of the 20-second clip — consistent with the onset of an arrhythmia. The correlation method pinpoints this moment precisely (beat-level resolution); the spectrogram corroborates the same transition with a visible breakdown in the harmonic banding, though with coarser time resolution, as expected from the time/frequency trade-off discussed throughout this report.